

Actinomycosis

Etiology

- **Filamentous:** Thread-like in structure.
- **Non-acid-fast:** Does not retain specific stains like acid-fast bacteria.
- **Non-spore-forming:** Lacks the ability to produce spores for survival.
- **Anaerobic:** Grows in the absence of oxygen.

Pathophysiology

- **Devitalized tissue:** Dead or damaged tissue.
- **Polymicrobial infection:** Involves multiple types of microorganisms.
- **Suppress host defenses:** Reduces the body's ability to fight infections.
- **Fibrosis:** Formation of excess fibrous connective tissue (scarring).
- **Contiguous spread:** Spreads directly to neighboring tissues.
- **Sinus tracts:** Channels or pathways from an internal infection to the surface.
- **Hematogenous spread:** Spread through the bloodstream.

Clinical Symptoms

A. Cervicofacial Actinomycosis

- **Submandibular/perimandibular:** Regions near or around the lower jaw.
- **Sulfur granules:** Yellowish particles in pus due to infection.
- **Trismus:** Difficulty or inability to open the mouth.
- **Mastication muscles:** Muscles used for chewing.
- **Lymphadenopathy:** Swelling of lymph nodes.

B. Thoracic Actinomycosis

- **Aspiration:** Inhalation of foreign material into the lungs.
- **Blood-streaked sputum:** Mucus coughed up with traces of blood.
- **Pleurocutaneous fistula:** Abnormal connection between the pleural cavity (chest) and the skin.
- **Systemic:** Affects the whole body rather than just one part.
- **Fatigue:** Extreme tiredness.
- **Anorexia:** Loss of appetite.

Abdominal Actinomycosis

History

- **Perforation:** A hole or break in the wall of an organ.

- **Foreign body ingestion:** Swallowing non-food items.

Symptoms

- **Perianal region:** Area around the anus.

Pelvic Actinomycosis

History

- **IUCD:** Intrauterine contraceptive device (a device used for birth control).

Symptoms

- **Vaginal discharge:** Fluid from the vagina, often due to infection.

Management

A. Diagnosis

1. **CBC:** Complete blood count (a blood test to evaluate overall health).
 - **Mild anemia:** Slight reduction in red blood cells or hemoglobin.
 - **Leucocytosis:** Increased white blood cell count.
2. **ESR, CRP:**
 - **ESR:** Erythrocyte sedimentation rate (measures inflammation).
 - **CRP:** C-reactive protein (a marker of inflammation).
3. **Culture:**
 - **Sulfur granules:** Yellow particles in pus due to infection.
 - **Beaded, branched rods:** Appearance of Actinomyces under the microscope.
4. **Imaging:**
 - **CT:** Computed tomography (advanced imaging technique).
 - **USG-guided aspiration:** Using ultrasound to guide the removal of fluid/tissue.
 - **Biopsy:** Sampling tissue for examination.
 - **Exploratory surgery:** Surgery performed to investigate an internal issue.

B. Treatment

1. Medications:

- **Penicillin G:** Antibiotic, main treatment.
- **Doxycycline:** An alternative antibiotic, often for penicillin-allergic patients.
- **Sulfonamides:** A group of antibiotics including sulfamethoxazole.

2. Surgery:

- **Drain abscesses:** Remove pus from infected areas.
- **Excise fibrotic lesions:** Cut out scarred tissue.
- **Relieve obstruction:** Remove blockages in affected areas.

Emergency Condition with Prolonged Skeletal Muscle Contraction

Etiology and Pathology

- **Clostridium tetani:** A type of bacteria causing tetanus.
- **Anaerobic:** Thrives without oxygen.
- **Spores:** Dormant and resistant forms of bacteria.
- **Tetanospasmin:** A neurotoxin causing muscle spasms.
- **Tetanolysin:** A toxin with unclear function.
- **Neurotransmitters:** Chemicals that transmit signals in the nervous system.

Clinical Features

- **Trismus:** Inability to open the mouth fully (lockjaw).
- **Dysphagia:** Difficulty swallowing.
- **Reflex spasms:** Involuntary muscle contractions due to stimuli.
- **Apnoea:** Temporary cessation of breathing.
- **Fractures:** Breaks in bones due to muscle spasms.
- **Rhabdomyolysis:** Breakdown of muscle tissue releasing harmful substances into the blood.
- **Laryngeal spasms:** Involuntary contractions of the voice box muscles.
- **Asphyxia:** Oxygen deprivation leading to suffocation.
- **Risus sardonicus:** Facial spasm causing a sneering grin.
- **Opisthotonus:** Severe arching of the body backward.
- **Autonomic dysfunction:** Abnormal control of blood pressure and heart rate.

Diagnostic Test

- **Spatula test:** A diagnostic test where the jaw reflexively bites down on a spatula.

Types of Tetanus

A. Generalized Tetanus

- **Descending pattern:** Symptoms begin in one area (e.g., jaw) and spread downward.

Neonatal Tetanus

- **Maternal immunity:** Protection passed from the mother to the baby through antibodies.
- **Umbilical stump:** The remaining part of the umbilical cord attached to the newborn's belly button.
- **Non-sterile instruments:** Tools not properly cleaned to kill germs.

Local Tetanus

- **Persistent:** Continuing for a long time without improvement.
- **Spasms:** Sudden, involuntary muscle contractions.

Cephalic Tetanus

- **Otitis media:** Infection of the middle ear.
- **Cranial nerves:** Nerves that originate in the brain, controlling functions like movement and sensation in the head.
- **CN VII:** The seventh cranial nerve, responsible for facial movement (facial nerve).

Management

A. Diagnosis

- **Clinical presentation:** Visible signs and symptoms observed in the patient.
- **Necrotic tissue:** Dead tissue caused by injury or infection.
- **Toxin detection:** Testing for poisonous substances produced by bacteria.

B. Treatment

- **Metronidazole:** An antibiotic used to treat bacterial infections.
- **Tetanus immunoglobulin (TIG):** A substance containing antibodies to neutralize tetanus toxin.
- **Muscle relaxants:** Medications to reduce muscle stiffness and spasms.
- **Diazepam:** A medication that calms the brain and muscles.
- **Beta-blockers:** Drugs that regulate heart rate and blood pressure.

Supportive Care

- **Airway:** The passage through which air travels to the lungs.
- **Vitals:** Essential signs of life, like heart rate, breathing rate, and blood pressure.

Lab Studies

- **Trismus:** Inability to open the mouth fully.
- **Dysphagia:** Difficulty swallowing.
- **Rigidity:** Stiffness of the muscles.

Spatula Test

- **Oropharynx:** The part of the throat at the back of the mouth.
- **Masseter:** The jaw muscle used for chewing.

Treatment

1. Wound Care

- **Thoroughly:** Completely and carefully.
- **Surgical debridement:** Removal of dead or infected tissue using surgery.
- **Necrotic:** Dead tissue due to injury or lack of blood supply.

2. Antibiotics

- **Metronidazole IV:** Intravenous antibiotic to treat bacterial infections.

3. Passive Immunization

- **Tetanus immunoglobulin (TIG):** Antibodies used to neutralize the tetanus toxin.
- **Unbound toxin:** Toxins not yet attached to nerve tissues.
- **IM:** Intramuscular (injection into muscle).
- **WHO:** World Health Organization, providing global health guidelines.
- **TT vaccine:** Tetanus toxoid vaccine, used to prevent tetanus.

Mild Tetanus

- **Diazepam:** Medication for calming spasms and anxiety.

Severe Tetanus

- **ICU:** Intensive Care Unit for critically ill patients.

- **Intrathecal TIG:** Tetanus immunoglobulin injected into the spinal canal.
- **Intubation:** Insertion of a tube into the airway to assist breathing.
- **Tracheostomy:** Surgical creation of an opening in the windpipe for breathing.
- **Reflex laryngospasm:** Sudden, involuntary closing of the vocal cords.
- **Succinylcholine:** A drug used to relax muscles quickly during intubation.
- **Magnesium IV infusion:** Intravenous magnesium to prevent spasms.
- **Curare-like drugs:** Medications causing temporary paralysis.
- **Mechanical ventilation:** A machine that helps a patient breathe.

Supportive Care

1. Nutrition

- **PEG tube:** Percutaneous endoscopic gastrostomy, a feeding tube placed into the stomach.
- **TPN (Total Parenteral Nutrition):** Nutrition provided intravenously when oral feeding isn't possible.

2. Airway Maintenance

- **Critical:** Essential for survival.

3. Recovery

- **Regenerate:** To grow or repair tissue.

Carbuncle

Definition

- **Carbuncle:** A deep skin infection involving multiple hair follicles.
- **Confluence:** Joining or merging of parts into one.
- **Furuncles:** Skin boils, infections of single hair follicles.
- **Abscess:** A swollen area filled with pus.

Aetiology

- **Staphylococcus aureus:** A common bacteria causing infections.
- **Contagious:** Easily spread from person to person or to other areas.
- **Dermatitis:** Inflammation of the skin.

Clinical Features

- **Cluster:** A group or collection of similar items close together.
- **Nape:** The back part of the neck.
- **Crust:** A hardened layer forming on the skin.
- **Fatigue:** Extreme tiredness.

Treatment

Self-Care

- **Compress:** A soft, moist cloth applied to the skin for relief.

Medical Treatment

- **Antibacterial soap:** Soap designed to kill bacteria.
- **Surgical drainage:** A procedure to remove pus from an infected area.
- **Cruciate incision:** A cross-shaped surgical cut.

Prevention of Spread

- **Bandages:** Protective coverings for wounds.
- **Dispose:** To throw away safely.
- **Suppress:** Reduce or control something undesirable.

Definition

- **Clostridial Myonecrosis:** A severe bacterial infection of muscle tissue.
- **Gangrene:** Death of body tissue due to loss of blood supply or infection.
- **Clostridium perfringens:** A toxin-producing bacterium causing the infection.

Pathogenesis

- **Anaerobic:** Surviving without oxygen.
- **Gram-positive:** Bacteria with a thick cell wall that retains a violet stain during Gram staining.
- **Spore-forming:** Bacteria capable of producing spores to survive harsh conditions.
- **Alpha-toxin:** A toxic protein damaging cells.
- **Lecithinase (phospholipase-C):** An enzyme breaking down cell membranes.
- **Necrosis:** Death of cells or tissues.
- **Oedema:** Swelling caused by fluid accumulation.
- **Thrombosis:** Formation of blood clots inside vessels.
- **Fermentation:** Breakdown of substances like glucose by bacteria, producing gas.

- **Hemolysis:** Destruction of red blood cells.
- **Renal failure:** Loss of kidney function.
- **Shock:** A critical condition due to inadequate blood flow to tissues.

Clinical Features

- **Trauma:** Physical injury.
- **Occult malignancy:** Hidden or undiagnosed cancer.
- **Serosanguineous exudate:** Fluid mixed with blood and serum from a wound.
- **Bronze:** A brownish color change in the skin.
- **Bullae:** Large fluid-filled blisters.
- **Blebs:** Small fluid-filled sacs on the skin.
- **Crepitus:** Crackling sensation under the skin due to gas.
- **Sweet/mousy odor:** A distinctive smell sometimes associated with infection.

Late Signs

- **Hypotension:** Low blood pressure.
- **Mental clarity paradox:** Clear mental state despite severe illness.
- **Tachycardia:** Abnormally fast heart rate.
- **Altered mental state:** Confusion or other changes in awareness.

Management

1. Surgical Treatment

- **Debridement:** Surgical removal of dead or infected tissue.
- **Amputation:** Surgical removal of a limb or part of the body.
- **Fasciotomy:** A surgical procedure to relieve pressure in muscle compartments.

2. Antibiotic Therapy

- **Penicillin G:** A broad-spectrum antibiotic effective against gram-positive bacteria.
- **Clindamycin:** An antibiotic that inhibits bacterial protein synthesis.
- **Metronidazole:** An antibiotic targeting anaerobic bacteria.

3. Hyperbaric Oxygen Therapy (HBO)

- **Hyperbaric:** Involving high-pressure oxygen.
- **Adjunct:** An additional treatment used alongside primary therapies.

Summary of Treatment

- **Prompt:** Quick and timely.
- **Adjunctive:** Supplementary or secondary to the main treatment.
- **Clinical evidence:** Research data supporting medical practices.

Alveolar Abscess (Dentoalveolar Abscess)

Definition

- **Abscess:** A collection of pus caused by infection.
- **Alveolar ridge:** The bony ridge in the jaw that holds teeth.
- **Nonvital tooth:** A tooth that has lost its blood supply, typically due to decay or trauma.
- **Suppurative inflammation:** Inflammation with pus formation.
- **Root apex:** The tip of the root of a tooth.

Pathology

1. **Periapical Abscess:**
 - **Dental pulp:** The innermost part of the tooth containing nerves and blood vessels.
 - **Pulpitis:** Inflammation of the dental pulp.
 - **Necrosis:** Death of tissue.
2. **Periodontal Abscess:**
 - **Periodontal ligaments:** Tissues that support and anchor the teeth in the jawbone.
 - **Alveolar bone:** The bone that surrounds the roots of the teeth.
3. **Pericoronitis:**
 - **Gum flap:** Tissue that partially covers a tooth, often in the case of a partially erupted tooth.
 - **Erupted:** When a tooth breaks through the gum line.

Clinical Features

1. **Local Symptoms**
 - **Localized:** Confined to a specific area.
 - **Thermal sensitivity:** Sensitivity to hot or cold due to dentin exposure.
 - **Gingival bleeding:** Bleeding from the gums.
 - **Reduced intake:** Difficulty eating or drinking.
2. **Physical Signs**
 - i. **Gingiva:**
 - **Erythema:** Redness due to increased blood flow.
 - **Fluctuance:** The ability to feel a fluid-filled area that moves under pressure.

- **Parulis ("gum boil"):** A small, swollen area on the gum with a drainage point.

ii. Teeth:

- **Tenderness:** Pain or discomfort when pressure is applied.
- **Mobility:** Movement of the tooth in its socket.
- **Extrusion:** The tooth being pushed out of its socket.

iii. Regional Lymph Nodes:

- **Involved:** Swollen or painful due to infection.

iv. Severe Infection:

- **Trismus:** Difficulty opening the mouth due to muscle spasm, often indicating deep tissue involvement.
- **Dysphagia:** Difficulty swallowing.
- **Necrotizing fasciitis:** A severe, rapidly spreading soft tissue infection.
- **Swelling:** Enlarged or puffed-up appearance, often in the face or neck.

Treatment

1. Airway Management

- **Respiratory distress:** Difficulty in breathing.
- **Oropharyngeal swelling:** Swelling in the mouth and throat area.
- **Intubation:** Inserting a tube into the airway to assist with breathing.
- **Tracheostomy:** Surgical procedure to create an opening in the trachea (windpipe) to help with breathing.

2. Diagnosis

- **Specimens:** Samples taken from the infected area to identify the bacteria.
- **Gram stain:** A laboratory technique used to identify bacteria by their cell wall structure.
- **Culture:** Growing bacteria from a sample to identify the infection-causing organism.
- **Aerobic/anaerobic:** Refers to whether bacteria need oxygen (aerobic) or can survive without it (anaerobic).

3. Medications

- **Antibiotics:** Medications used to treat bacterial infections.
- **Empiric therapy:** Initial treatment given before the specific cause of infection is identified.

- **Analgesics:** Pain-relieving medications.
- **Hydration:** Ensuring adequate fluid intake to maintain body functions.

4. Procedures

- **Pulpectomy:** Removal of the pulp (nerve and blood supply) from a tooth.
- **Incision and drainage:** Surgical procedure to open and drain an abscess.
- **Tooth extraction:** Removal of a tooth from the socket.

Ludwig's Angina

1. Cellulitis:

- A bacterial infection causing redness, swelling, and inflammation of the skin and underlying tissues.

2. Submental Region:

- The area beneath the chin, between the mandible and hyoid bone.

3. Submandibular Region:

- The area below the jaw, near the submandibular glands.

4. Streptococcal Organisms:

- A type of bacteria that can cause various infections, including dental and throat infections.

5. Edema:

- Swelling caused by fluid accumulation in tissues.

6. Halitosis:

- Foul-smelling breath, often due to infection or poor oral hygiene.

7. Induration:

- Hardening of tissues due to infection or inflammation.

8. Trismus:

- Difficulty in opening the mouth due to jaw muscle spasm.

9. Dyspnea:

- Difficulty breathing, often caused by airway obstruction.

10. Glottic Edema:

- Swelling of the vocal cords, leading to breathing problems.

11. Ryle's Tube:

- A nasogastric tube inserted through the nose into the stomach for feeding.

12. Septicemia:

- A serious condition where bacteria spread into the bloodstream, causing systemic infection.

13. Mediastinitis:

- Inflammation or infection of the mediastinum, the central part of the chest cavity.

14. Tracheostomy:

- A surgical procedure to create an opening in the windpipe (trachea) for breathing.

Erysipelas

Definition

- **Acute:** Sudden onset.
- **Superficial:** Affects the outer layers of the skin.
- **Streptococcus bacterial infection:** Infection caused by bacteria from the *Streptococcus* genus.
- **Epidermis:** The outermost layer of skin.
- **Cutaneous lymphatic spread:** Spread of infection through the lymphatic vessels in the skin.

Aetiology and Pathology

1. Causes

- **Streptococcus pyogenes:** A type of bacteria that causes infections like erysipelas.
- **Beta-hemolytic group A streptococci:** A classification of *Streptococcus pyogenes* based on its ability to break down red blood cells.
- **Non-group A streptococci:** Other types of *Streptococcus* that are not group A but can still cause infection.

2. Development

- **Bacterial inoculation:** Introduction of bacteria into the body, typically through a break in the skin.
- **Venous insufficiency:** Poor blood flow in the veins, often causing swelling and skin issues.
- **Stasis ulcerations:** Ulcers caused by poor circulation.
- **Inflammatory dermatoses:** Skin conditions that cause inflammation, such as eczema.
- **Dermatophyte infections:** Fungal infections of the skin.
- **Insect bites:** Bites from insects, which can introduce bacteria into the skin.
- **Surgical incisions:** Cuts made during surgery that may become infected.
- **Lymphatic vessels:** Blood vessels that carry lymph (fluid containing immune cells).
- **Skin "streaking":** Visible lines of infection spreading through the skin, often along lymphatic vessels.
- **Lymph node swelling:** Enlargement of the lymph nodes due to infection.

Clinical Features

1. General Symptoms

- **High fever:** Elevated body temperature, often a sign of infection.
- **Shaking chills:** Sudden coldness and shaking, often with fever.
- **Fatigue:** Extreme tiredness.
- **Headaches:** Pain in the head area.
- **Vomiting:** Throwing up.
- **General feeling of illness:** Unwell sensation that can include aches, fatigue, and fever.

2. Skin Symptoms

- **Erythematous:** Red, inflamed skin.
- **Indurated:** Hardened or firm skin.
- **Tense:** Skin feels tight.
- **Shiny plaque:** Raised, shiny area of skin that is typically red or inflamed.
- **Lesion:** Abnormal area of tissue.
- **Sharp demarcation:** Clear and distinct border between infected and healthy skin.
- **Vesicles:** Small fluid-filled blisters.
- **Bullae:** Larger fluid-filled blisters.
- **Petechiae:** Small, red or purple spots caused by bleeding under the skin.
- **Skin necrosis:** Death of skin tissue, leading to decay or blackening.

3. Lymphatic Involvement

- **Lymphadenopathy:** Swelling of the lymph nodes.
- **Tender lymph nodes:** Lymph nodes that are painful to the touch.

4. Common Locations

- **Extremities:** Limbs, including arms and legs.
- **Facial involvement:** Infection affecting the face, especially around the eyes, ears, and cheeks.
- **Fat tissue:** Tissue beneath the skin, which is more prone to infection.

Treatment

1. Supportive Care

- **Elevation:** Raising the affected limb to reduce swelling.
- **Rest:** Allowing the body to heal.
- **Saline wet dressings:** Damp cloths soaked in saltwater to clean and soothe the skin.
- **Necrotic lesions:** Areas of skin that have died due to infection.

2. Medications

- **Penicillin:** A type of antibiotic used to treat bacterial infections.
- **Oral or intramuscular:** Methods of administering penicillin, either by mouth or injection.
- **Therapy duration (10-20 days):** Length of time the antibiotic is taken to ensure the infection is cleared.

Boil (or Furuncle)

Definition

- **Deep folliculitis:** Infection of the hair follicle beneath the skin's surface.
- **Staphylococcus aureus:** A type of bacteria commonly responsible for causing boils.
- **Pus:** A thick fluid produced by the body during infection, consisting of dead cells and bacteria.
- **Dead tissue:** Tissue that has died due to infection or lack of blood supply.
- **Carbuncle:** A cluster of interconnected boils.

Clinical Features

1. Appearance

- **Bumpy:** Raised, irregular shape.
- **Pus-filled:** Containing pus, the thick fluid produced during infection.
- **Lumps:** Swollen, raised areas of skin.
- **Tender:** Painful to the touch.
- **Warm:** The area feels hot due to inflammation.
- **Yellow or white point:** The center of the boil, indicating it is ready to drain.

2. Systemic Symptoms

- **Fever:** Elevated body temperature as part of the body's immune response.
- **Swollen lymph nodes:** Lymph nodes, which help fight infection, become enlarged and tender when battling infection.
- **Fatigue:** Extreme tiredness as the body fights infection.

3. Risk Factors

- **Systemic factors:** Conditions affecting the entire body, not just localized areas.
- **Lower immunity:** A weakened immune system increases susceptibility to infections.
- **Diabetes:** A condition that affects blood sugar levels and immune function.
- **Obesity:** Excess body fat that can affect immune function.
- **Hematologic disorders:** Conditions related to the blood, which may impact the immune response.

Treatment

1. Home Care

- **Warm salt water compresses:** A warm cloth soaked in salt water used to encourage the boil to drain.
- **Antibiotic cream:** A topical medication to kill bacteria and prevent further infection.
- **Antiseptic tea tree oil:** An essential oil with antibacterial properties that can be applied to the boil to reduce infection risk.
- **Bandage:** A covering applied over the treated boil to protect it and encourage healing.

2. Medical Supervision

- **Never squeeze or lance:** Do not attempt to open or drain the boil without medical supervision to avoid spreading the infection.

3. Antibiotics

- **Antibiotic therapy:** Medication used to treat bacterial infections.
- **Large or recurrent boils:** Boils that are particularly large or that keep coming back may require antibiotics.
- **Sensitive areas:** Boils occurring in delicate areas like the face, groin, or armpits may need medical treatment due to their location.

Syphilis

Definition

- **Venereal disease:** A sexually transmitted disease (STD).
- **Spirochete:** A type of bacteria with a spiral shape.
- **Treponema pallidum:** The bacterium responsible for causing syphilis.
- **Transmitted:** Spread from one person to another.

Stages of Syphilis

1. Primary Syphilis

- **Papule:** A small, raised bump on the skin.
- **Glans penis:** The head of the penis.
- **Vulva:** The external part of the female genitalia.
- **Cervix:** The lower part of the uterus.
- **Chancre:** A painless ulcer or sore.
- **Ulcerates:** Develops into an open sore.
- **Crater:** A bowl-shaped depression.
- **Elevated edges:** Raised borders around the ulcer.

2. Secondary Syphilis

- **Mucous membrane:** The lining of various body cavities, such as the mouth and genital tract.
- **Lesions:** Abnormal areas of tissue, such as sores or rashes.
- **Condyloma latum:** A type of wart-like lesion found in syphilis.
- **Lymphadenopathy:** Swollen lymph nodes.
- **Maculopapular:** A rash that has both flat (macules) and raised (papules) areas.
- **Pustular:** Containing pus.

3. Latent Syphilis

- **Asymptomatic:** Showing no symptoms.

4. Tertiary Syphilis

- **Progressive:** Gradually worsening or advancing.
- **Gumma:** Soft, tumor-like growths that occur in tertiary syphilis.
- **Bone pain:** Pain in the bones, often described as deep and worse at night.
- **CNS (Central Nervous System):** The brain and spinal cord.

Diagnosis

1. Blood Tests

- **Nontreponemal tests:** Initial tests to detect antibodies produced in response to syphilis.
- **VDRL:** Venereal Disease Research Laboratory test, a nontreponemal test.
- **Rapid plasma regain:** Another nontreponemal test.
- **Treponemal tests:** Confirmatory tests that specifically detect the presence of *Treponema pallidum*.
- **TPHA:** *Treponema pallidum* particle agglutination test, a confirmatory test.
- **FTA-Abs:** Fluorescent treponemal antibody absorption test, a confirmatory test.

Treatment

1. Penicillin

- **Mainstay:** The primary or standard treatment.
- **Benzathine penicillin G:** A long-acting form of penicillin used to treat syphilis.
- **IM (Intramuscular):** Administered by injection into the muscle.
- **Units:** The measurement used to determine the dose of medication.

Pregnancy

- Treatment may vary depending on the stage of syphilis during pregnancy.

HIV (Human Immunodeficiency Virus) and AIDS (Acquired Immunodeficiency Syndrome)

Definition

- **HIV:** A virus that attacks the immune system, specifically CD4 cells, making the body more vulnerable to infections.
- **AIDS:** The final stage of HIV infection, where the immune system is severely weakened.

Precautions in OPD (Outpatient Department)

1. Gloves

- **Gloves:** Protective coverings worn on hands to prevent direct contact with bodily fluids.

2. Disposable Instruments

- **Disposable instruments:** Medical tools that are used once and then discarded to avoid contamination.

3. Reusable Instruments

- **Reusable instruments:** Tools that are sterilized and reused in multiple procedures.
- **Glutaraldehyde:** A chemical disinfectant used to sterilize medical instruments.

Precautions in Operating Room

1. Dental Chair

- **Polythene:** A type of plastic material used to cover surfaces for hygiene purposes.

2. Personnel

- **Personnel:** The staff members involved in medical procedures.
- **Abrasions:** Scrapes or shallow cuts in the skin.
- **Lacerations:** Deep cuts or tears in the skin.

3. Surgical Technique

- **Needle-stick injuries:** Injuries caused by a needle accidentally pricking the skin.

4. Post-Surgery

- **Autoclaving:** A process of sterilizing equipment by subjecting it to high-pressure steam.

Post-Exposure Prophylaxis for Health Workers

- **AZT (Zidovudine):** An antiretroviral medication used to prevent HIV infection.
- **Lamivudine:** An antiretroviral drug used to treat HIV/AIDS.
- **Indinavir:** A protease inhibitor used in the treatment of HIV/AIDS.

Causes of Hyperkalaemia

1. Impaired Excretion

- **Acute renal failure:** Sudden loss of kidney function, leading to the accumulation of potassium in the blood.
- **Chronic renal failure (CRF):** Long-term kidney dysfunction causing potassium retention.
- **Addison's disease:** A condition where the adrenal glands do not produce enough hormones, leading to potassium retention.
- **Hypoaldosteronism:** A condition where the body produces insufficient aldosterone, causing potassium retention.
- **Potassium-sparing diuretics:** Medications that reduce the amount of water in the body without losing potassium.

2. Excessive Intake

- **Intravenous fluids containing K⁺:** Fluids infused into the body that contain potassium, which may raise potassium levels.
- **High K⁺ foods:** Foods rich in potassium that can contribute to elevated potassium levels.

3. Tissue Breakdown

- **Haemolysis:** The destruction of red blood cells, releasing potassium into the bloodstream.
- **Rhabdomyolysis:** Breakdown of muscle tissue, releasing potassium into the blood.
- **Crush injury:** Physical injury to muscle tissue that can lead to the release of potassium into the bloodstream.

4. Shift of K⁺ Out of Cell

- **Acidosis:** An increase in the acidity of the blood, causing potassium to shift from inside cells to the bloodstream.
- **Insulin deficiency:** Insulin helps move potassium into cells; lack of insulin can lead to high potassium levels in the blood.

5. Pseudohyperkalaemia

- **Haemolysed blood sample:** When blood cells break down during or after sample collection, falsely increasing potassium levels in the test result.

Clinical Features

- **Cardiac arrhythmias:** Irregular heartbeats, which can be life-threatening.
- **Muscular weakness:** Weakness in muscles, which may progress to flaccid paralysis (loss of muscle tone).
- **Respiratory embarrassment:** Difficulty breathing due to muscle weakness affecting the respiratory muscles.

Electrocardiographic Manifestations of Hyperkalaemia

- **Tall peaked T waves:** A sign on the ECG that indicates elevated potassium levels.
- **Prolongation of PR interval:** A delayed signal transmission in the heart due to elevated potassium.
- **Reduced height of P wave:** Decrease in the P wave amplitude on ECG due to hyperkalaemia.
- **Prolongation of QRS complex:** Widening of the QRS complex on ECG, indicating delayed ventricular contraction.
- **Sine wave pattern:** A dangerous ECG pattern indicating severe hyperkalaemia.
- **Ventricular fibrillation:** A life-threatening arrhythmia that may occur due to very high potassium levels.
- **Ventricular standstill:** The heart may stop beating altogether due to elevated potassium levels.

Management

1. **Identification and elimination of the underlying cause:** Address the root cause of hyperkalaemia to prevent recurrence.
2. **If marked ECG changes, give:**
 - **Calcium gluconate solution:** 10 ml of 10% calcium gluconate intravenously to stabilize the heart's electrical activity.
3. **Shift of potassium into cells:**
 - **Intravenous glucose with insulin:** A combination of glucose and insulin helps lower blood potassium by shifting it into cells.
4. **Correct acidosis:**
 - **Sodium bicarbonate:** A substance used to correct acidosis and shift potassium back into cells.
5. **Nebulisation of beta-agonists:**
 - **Salbutamol or terbutaline:** Medications that can lower potassium levels by promoting the movement of potassium into cells.
6. **Cation exchange resins:**
 - **Sodium polystyrene sulphonate:** A resin used to help remove potassium from the body, administered orally or via enema.

7. **If above measures fail, haemodialysis may be needed:** In severe cases, dialysis may be required to remove excess potassium from the blood.

Swelling of Jaw in Osteomyelitis:

- **Definition:** Osteomyelitis of the jaw is an infection of the bone, often caused by bacteria, leading to swelling, pain, and tenderness.
- **Cause:** Most commonly caused by **Staphylococcus aureus** or **Streptococcus** following dental infections, trauma, or surgical procedures.
- **Symptoms:**
 - **Swelling** of the jaw.
 - **Pain** and tenderness.
 - **Redness** and warmth over the affected area.
 - **Fever** and malaise may occur in severe cases.
 - **Tooth mobility** and **abscess** formation.
 - **Pus** or **fistula** may drain from the affected area.

Investigations:

- **Clinical Examination:** Tenderness, swelling, and redness over the affected jaw.
- **X-ray:** Helps to detect bone destruction and **radiolucent areas** indicating infection.
- **CT Scan/MRI:** Provides detailed imaging of bone and soft tissue involvement.
- **Blood Tests:**
 - Elevated **WBC count**.
 - Raised **ESR** and **CRP** indicating inflammation.
- **Microbiological Culture:** For identification of causative organism from pus or blood.

Treatment:

- **Antibiotics:**
 - Empirical broad-spectrum antibiotics until culture results are available.
 - Once the organism is identified, use specific antibiotics (e.g., **penicillin**, **clindamycin**).
- **Surgical Drainage:**
 - If abscess or pus formation is present, surgical drainage is necessary to remove infected material.
- **Pain Management:**
 - Use of **analgesics** like ibuprofen or paracetamol.
- **Supportive Care:**
 - Hydration and nutritional support.
- **Root Canal Treatment/Extraction:** If the infection is related to a tooth, root canal treatment or tooth extraction may be necessary.

- **Hyperbaric Oxygen Therapy:** In severe or chronic cases, to promote healing and enhance antibiotic effectiveness.

Prognosis: Early treatment results in a good prognosis, while delays may lead to chronic osteomyelitis or complications like **sepsis**.

Chronic osteomyelitis

- **Definition:** It is an infection in the bone, leading to a complex inflammatory process.

Complex Inflammatory Process:

- **Inflammation:** Body's response to injury or infection, causing redness, swelling, and pain.

Necrosis of Mineralized and Marrow Tissues:

- **Necrosis:** Death of cells or tissues.
- **Mineralized Tissues:** Tissues that have hardened due to deposition of minerals like calcium, such as bone.
- **Marrow Tissues:** Soft tissue inside bones that produces blood cells.

Suppuration:

- **Suppuration:** The process of pus formation during infection.

Resorption:

- **Resorption:** The process where the bone or other tissues are broken down and absorbed back into the body.

Sclerosis:

- **Sclerosis:** Hardening of tissues, often due to excessive deposition of collagen or minerals.

Hyperplasia:

- **Hyperplasia:** Increase in the number of cells in an organ or tissue, leading to its enlargement.

Pathogenesis

- **Infection of bone marrow from infected pulp:** Bacterial infection spreads from the infected tooth pulp into the bone marrow of the jaw.

- **Extension of infection into cancellous bone:** The infection spreads from the bone marrow into the spongy, inner part of the bone.
- **Thrombus formation in nutrient vessels of the bone:** Formation of blood clots in the vessels supplying the bone, reducing blood flow.
- **Death of cancellous bony trabeculae, leading to formation of sequestrum:** The trabeculae (small bony structures) of the spongy bone die, creating a dead bone fragment (sequestrum).
- **Spread of infection via Volkmann's canal in cortical plates:** The infection spreads through small channels (Volkmann's canals) in the outer layer (cortical plate) of bone.
- **Periostitis (inflammation of the periosteum):** Inflammation of the periosteum, the connective tissue surrounding the bone.
- **Multiple sinus tract formation:** Formation of multiple channels or openings in the bone or soft tissue, often leading to pus discharge.
- **Necrosis of cortical bone:** Death of the outer layer (cortical bone) of the jaw bone.
- **Discharge of pus through sinuses (called cloacae):** Pus drains out of the body through openings called cloacae formed by the sinuses.

Clinical Examination:

- **Tenderness:** Sensitivity or pain when touched.
- **Swelling:** Enlargement or puffiness of a body part.
- **Redness:** Increase in blood flow causing a reddish appearance in the affected area.

X-ray:

- **Bone Destruction:** Breakdown or damage of bone tissue.
- **Radiolucent Areas:** Areas that appear dark on an X-ray due to less mineral content (indicating infection or bone loss).

CT Scan/MRI:

- **CT Scan:** Imaging technique using X-rays to create detailed cross-sectional images of the body.
- **MRI:** Magnetic Resonance Imaging, a technique that uses magnetic fields and radio waves to create detailed images of organs and tissues.

Blood Tests:

- **WBC (White Blood Cell) Count:** The number of white blood cells in the blood, indicating immune system activity.
- **ESR (Erythrocyte Sedimentation Rate):** A blood test measuring the rate at which red blood cells settle in a test tube, indicating inflammation.

- **CRP (C-Reactive Protein):** A protein in the blood that rises in response to inflammation.

Microbiological Culture:

- **Microbiological Culture:** A lab test that grows microorganisms (bacteria, fungi) from samples to identify the infecting organism.

Antibiotics:

- **Empirical:** Initial treatment based on likely causes before confirming the exact cause.
- **Broad-Spectrum Antibiotics:** Antibiotics effective against a wide range of bacteria.
- **Penicillin:** A type of antibiotic used to treat infections.
- **Clindamycin:** An antibiotic used for bacterial infections, often for bone and soft tissue infections.

Surgical Drainage:

- **Abscess:** A collection of pus caused by infection.
- **Surgical Drainage:** A procedure to remove pus or fluid from an infected area.

Pain Management:

- **Analgesics:** Medications used to relieve pain (e.g., ibuprofen, paracetamol).

Supportive Care:

- **Hydration:** Providing fluids to the body to maintain normal function.
- **Nutritional Support:** Ensuring the patient receives the necessary nutrients to aid recovery.

Root Canal Treatment/Extraction:

- **Root Canal Treatment:** A dental procedure to treat infection inside a tooth's pulp (inner tissue).
- **Extraction:** The removal of a tooth.

Hyperbaric Oxygen Therapy:

- **Hyperbaric Oxygen Therapy:** A treatment where the patient breathes pure oxygen in a pressurized room to promote healing and fight infection.

Clinical Features

- **Molar area of mandible is more commonly affected:** The infection often involves the back part of the lower jaw (mandible), where the molars are located.
- **Pain is usually mild and insidious, not related to disease severity:** Pain may not be intense and develops slowly, not necessarily correlating with the extent of the infection.
- **Jaw swelling is common, but mobility of teeth and sinus tract formation are rare:** Swelling of the jaw is common, but the teeth usually don't become loose, and sinus tracts are not frequently seen.
- **Anaesthesia and paraesthesia of lip are uncommon:** Numbness or abnormal sensations in the lip are rare.
- **Regional lymphadenopathy is common:** Swelling of lymph nodes near the infection site is common.
- **Thickened, wooden feeling of bone with slow increase in jaw size:** The bone feels thick and hard, like wood, with gradual growth in the size of the jaw.

Histopathology

- **Chronic inflammatory reaction of bone with accumulation of exudate and pus within medullary spaces:** Long-term inflammation in the bone, with pus and fluid accumulation in the inner part of the bone (medullary spaces).
- **Inflammatory cells: Lymphocytes, macrophages, and plasma cells predominate:** The primary cells involved in the inflammatory response are lymphocytes, macrophages, and plasma cells.
- **Osteoblastic and osteoclastic cavities partially form with irregular bony trabeculae having reversal lines:** Cavities formed by bone-forming cells (osteoblasts) and bone-resorbing cells (osteoclasts), with irregular bone structure and distinct lines indicating areas of bone remodeling.
- **Sequestrum may develop in later stages:** A piece of dead bone (sequestrum) may form as the infection progresses.
- **Colonies of bacteria can be seen within the inflamed tissue:** Bacterial clusters are visible within the inflamed bone tissue.

Radiological Findings

- **Moth-eaten appearance with irregular, enlarged radiolucent areas separated by islands of normal bone:** X-rays may show irregularly shaped, hollow areas (radiolucent) in the bone, surrounded by healthy bone.
- **Sequestra: Irregular calcified areas separated from normal bone:** Areas of dead bone (sequestrum) that are calcified, appearing separate from the surrounding healthy bone.

- **External resorption of tooth roots and lamina dura becomes less apparent:** The outer layers of the tooth roots and the surrounding bone (lamina dura) are often less visible due to bone destruction.
- **Fistulous tracts may appear:** Channels (fistulas) may form, which can lead to drainage of pus from the infection site.

Cancrum Oris

- **Vincent's disease:** An acute, severe infection of the gums, causing ulcers and necrosis (tissue death).
- **Necrosis:** The death of cells or tissues in the body, often due to infection or lack of blood supply.
- **Sequestrates:** The process in which dead bone becomes separated from healthy bone and tissue.

Progression of Infection

- **Necrosis (tissue death):** The process where the tissues die due to infection or injury.
- **Sequestered:** When infected or dead bone becomes separated from the healthy bone and tissue around it.

Post-Surgical Effects

- **Scarring:** The formation of fibrous tissue after an injury or infection, leading to permanent marks on the skin or tissue.
- **Facial asymmetry:** Uneven appearance of the face due to tissue loss or abnormal healing after an infection or surgery.
- **Trismus:** Difficulty opening the mouth due to muscle stiffness or scarring, often following infection or surgery.

Causative Organism

- **Human Immunodeficiency Virus (HIV):** A virus that attacks and weakens the immune system, eventually leading to acquired immunodeficiency syndrome (AIDS).

Mode of Transmission

- **Unprotected sexual intercourse:** Sexual contact without barrier methods (like condoms), increasing the risk of transmitting HIV.

- **Mother to fetus:** HIV can be passed from an infected mother to her child during pregnancy, childbirth, or breastfeeding.
- **Contaminated needles:** Sharing needles (commonly among drug users) can transmit HIV through blood.
- **Contaminated blood transfusion:** Receiving HIV-infected blood through transfusion.

Clinical Features

- **Prolonged diarrhoea:** Ongoing, chronic diarrhoea, a common symptom in people with advanced HIV/AIDS.
- **Weight loss:** Unexplained, significant loss of body weight.
- **Night sweating:** Excessive sweating during sleep, common in HIV/AIDS patients.
- **Prolonged fever:** Persistent fever, often indicating an ongoing infection or immune response.

Oral Manifestations of AIDS

- **Candidiasis (oral thrush):** A fungal infection in the mouth causing white patches or sores.
- **Angular cheilitis:** Inflammation and cracks at the corners of the mouth.
- **Kaposi's sarcoma:** A cancer affecting the skin and mucous membranes, common in AIDS patients.
- **Hairy leukoplakia:** White, abnormal patches on the tongue, often linked to HIV infection.
- **Herpes simplex and herpes zoster infections:** Viral infections causing sores on the mouth and skin.
- **Ulcerations in the mouth and throat:** Painful sores or ulcers in the mouth and throat area.
- **Delayed wound healing:** Slower recovery of injuries due to the weakened immune system.

Treatment

- **AIDS:** A disease caused by HIV that severely weakens the immune system, making the body vulnerable to infections and certain cancers.
- **Antiviral drugs:** Medications that target viruses like HIV, preventing their replication and reducing symptoms.
- **Azithromidine:** An antiviral drug used to treat HIV infections.
- **Zidovudine:** An antiviral medication used to manage HIV, preventing the virus from replicating.
- **Reverse transcriptase:** An enzyme that HIV uses to replicate; inhibiting this enzyme helps control the virus.

Safe Sexual Contact

- **Condoms:** A barrier method used during sexual intercourse to prevent the transmission of HIV and other sexually transmitted infections (STIs).

Prevent Sharing of Needles

- **Sterilized needles:** Needles that have been properly cleaned or treated to kill any germs, viruses, or bacteria to prevent infection.

Safe Blood Transfusion

- **HIV testing before transfusion:** Screening of blood donations to detect the presence of HIV, ensuring that only safe, non-contaminated blood is used for transfusions.

HIV Testing and Counseling

- **HIV screening:** The process of testing individuals for the presence of HIV.
- **Counseling and education:** Providing information and guidance to individuals about HIV prevention, testing, and safe practices to reduce the risk of transmission.

Prevention of Mother-to-Child Transmission

- **Antiretroviral therapy (ART):** A treatment involving a combination of HIV medicines to control the virus in the body and prevent transmission to others, especially to unborn children.
- **C-section delivery:** A surgical procedure in which a baby is delivered through an incision in the mother's abdomen, used to reduce the risk of HIV transmission during childbirth.
- **Breastfeeding alternatives:** Methods such as formula feeding, recommended for HIV-positive mothers to avoid transmitting the virus through breast milk.

Post-Exposure Prophylaxis (PEP)

- **PEP treatment:** A preventive treatment that involves taking HIV medications for 28 days, started within 72 hours of potential exposure to the virus, to prevent HIV infection.

PrEP (Pre-exposure Prophylaxis)

- **PrEP:** A preventive treatment involving daily medication (HIV drugs) to reduce the risk of HIV infection for individuals who are at high risk of exposure to the virus.

The ELISA (Enzyme-Linked Immunosorbent Assay)

- **ELISA:** A laboratory technique used to detect antibodies in the blood, often used as the first test for HIV.
- **Antibodies:** Proteins produced by the immune system in response to an infection.
- **Pre-test counselling:** Guidance given to individuals before testing to explain the process, potential outcomes, and support options.
- **Post-test counselling:** Guidance given after the test to discuss the results and next steps.

Western Blot

- **Western Blot:** A laboratory test used to confirm the presence of HIV antibodies by detecting the specific proteins of the virus.
- **Antibody detection:** The process of identifying antibodies that have formed in response to the HIV virus.
- **Viral proteins:** Specific proteins that are part of the HIV virus.
- **Serum antibodies:** Antibodies found in the liquid portion of blood that help identify infections like HIV.

Rapid HIV Testing

- **Pre-test and post-test counselling:** Guidance and support provided before and after the test to help individuals understand the process and results.
- **Centrifuged:** A process where blood is spun at high speed to separate its components, such as serum or plasma.

Types of Rapid HIV Tests:

1. **OraQuick/OraQuick Advanced Rapid HIV-1/2 Antibody Test**
 - **OraQuick:** A brand of rapid HIV test that can detect both HIV-1 and HIV-2 from blood or oral fluids.
 - **Venous blood:** Blood drawn from a vein.
 - **Plasma:** The liquid part of blood after the cells have been removed.
 - **Oral fluids:** Fluid from the mouth, including saliva.
2. **Reveal G2 HIV-1 Antibody Test**
 - **Reveal G2:** A rapid test for detecting HIV antibodies in plasma or serum.

- **Plasma or serum specimens:** Blood samples from which the cells have been removed, containing antibodies and other substances.
- **Centrifuged serum or plasma:** Blood components separated by spinning in a centrifuge.
- **Red dot:** A visual indication that HIV antibodies are present.
- 3. **Uni-Gold Recombigen HIV-1 Test**
 - **Uni-Gold:** A rapid test that detects HIV-1 in whole blood, plasma, or serum.
 - **Venipuncture:** The process of drawing blood from a vein.
 - **Finger stick:** A method of blood collection from a fingertip.
- 4. **Multispot HIV-1/HIV-2 Rapid Test**
 - **Multispot:** A type of rapid test that detects HIV-1 and HIV-2 in blood, plasma, or serum.
 - **Test cartridge:** A small device used for performing the test.
 - **Reagents:** Substances used in the test to detect HIV.

Aetiology (in Dental Clinic)

- **Infected instruments:** Dental tools that have been contaminated with HIV-infected blood or saliva.
- **Infected needles:** Needles that have been exposed to HIV-positive blood and may transmit the virus if reused.
- **Infected blood transfusion:** Receiving blood that is contaminated with HIV during dental procedures.
- **Contaminated gloves and dressing materials:** Protective gloves or materials used during dental procedures that are contaminated with HIV-infected blood or fluids.

Epidemiology

- **AIDS:** Acquired Immunodeficiency Syndrome, the final stage of HIV infection, characterized by severe immune system damage.
- **Globally:** Refers to the worldwide spread of a condition.
- **Sexually active adults:** Adults who are involved in sexual activity.
- **HIV-positive:** Refers to individuals who have been diagnosed with HIV, the virus that causes AIDS.

Pathology

- **CD4+ T cells:** A type of white blood cell crucial for the immune response, which HIV attacks and depletes.
- **Selective tropism:** The preference of HIV to infect specific cells, such as CD4+ T cells.
- **Internalization of HIV:** The process by which HIV enters cells.

- **Uncoating:** The process by which the viral coat is removed after HIV enters the host cell.
- **Proviral DNA integration:** The insertion of HIV's genetic material into the DNA of the infected cell.
- **Budding and syncytia formation:** The process by which new viral particles are formed and the fusion of infected cells into a larger cell.
- **Cytopathic effects:** Damage caused to cells by the virus, leading to cell death.
- **Monocytes and macrophages:** Types of white blood cells involved in the immune response, which can also be infected by HIV.
- **Nervous system:** The brain, spinal cord, and nerves, which can be affected by HIV leading to neurological complications.
- **B cell dysfunction:** The impairment of B cells, which are essential for producing antibodies and fighting infections.

Precautions in Dental Clinic

- **Sterilize instruments:** The process of cleaning dental instruments to eliminate any microorganisms, including HIV, to prevent transmission.
- **ELISA testing:** Enzyme-linked immunosorbent assay, a test used to detect HIV antibodies before blood transfusions.
- **Aseptic measures:** Techniques used to prevent infection, such as proper handwashing, sterilizing equipment, and using sterile gloves during surgery.
- **Educate the patient about AIDS:** Providing information to patients about HIV and AIDS to prevent transmission and promote awareness.